

XPLAIN™: EXPLAINABLE COST
SENSITIVE MACHINE LEARNING FOR AML
TRANSACTION MONITORING: REDUCING
FALSE POSITIVES AS COMPARED WITH
RULES-BASED SYSTEMS

MSC FINTECH (MSCFTD1)
RESEARCH IN COMPUTING - PRACTICUM CA1
PROFESSOR. FAITHFUL CHIAGOZIEM ONWUEGBUCHE

Mohd Nizam Mohd Nasir Shaikh

Student ID: 24198170

Student Outlook: X24198170@student.ncirl.ie

Academic Integrity Code: SfYu3HrPhR

School of Computing
National College of IRELAND

National College of Ireland

Project Submission Sheet

Student Name: Mohd Nizam Mohd Nasir Shaikh

Student ID: 24198170

Programme: Master of Science in FinTech MSCFTD1 Year: 2026-2027

Module: Research in Computing - PRACTICUM CA1

Lecturer: Prof. Faithful Chiagoziem Onwuegbuche

Submission

Due Date: 13/February/2025

Academic

Integrity Code: SfYu3HrPhR Word Count: 1018

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

ALL internet material must be referenced in the references section. Students are encouraged to use the Harvard Referencing Standard supplied by the library. To use other author's written or electronic work is illegal (plagiarism) and may result in disciplinary action. Students may be required to undergo a viva (oral examination) if there is suspicion about the validity of their submitted work.

Signature: Mohd Nizam

Date: 13/02/2026

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4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
5. All projects must be submitted and passed in order to successfully complete the year. **Any project/assignment not submitted will be marked as a failure.**

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AI Acknowledgement Supplement:

Module: Research in Computing - PRACTICUM CA1 (MSCFTD1)

Your Name/Student Number	Course	Date
Mohd Nizam Shaikh / 24198170	Master of Science in FinTech	13/02/2026

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment:

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
N/A	N/A	N/A

Description of AI Usage:

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

Evidence of AI Usage:

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

[N/A]

Academic Honesty Declaration

I declare the following to be true for this submission:

- I have completed the task during the designated time window and declare it to be exclusively my own work.
- I have not received, or attempted to receive assistance in preparing this response from any other person during the assessment window.
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TITLE: XplainTM - Explainable cost sensitive Machine Learning for AML transaction monitoring: Lowering false positives as compared with rules-based systems

Keywords: RegTech; AML transactions monitoring; imbalanced classification; cost sensitive learning; explainable AI (XAI); model evaluation and calibration

1. Research Problem Background

For financial firms, there is always a constant pressure to comply with Anti-Money Laundering (AML) related requirements without making compliance operations efficient (Oztas et al., 2024). Regtech supports this through the use of technology to improve the traditional way of monitoring, investigation and reporting processes are conducted in practice (Arner, Barberis and Buckley, 2017).

A relatively well-known weakness with traditional AML transaction monitoring is their reliance on static rules and thresholds (Oztas et al., 2024). While rules are easy to implement, they tend to produce huge numbers of false positives. This increases work load for analyst although slow down investigations and cut the amount of attention available for truly suspicious cases.

Recent research is pointing towards potential of machine learning to give enhanced patterns in detecting based on learning of complex patterns of the learner that rules struggle to capture. However, for ML to be realistic in regulatory environments it should be able to audited as well. On the other hand, it's not sufficient to generate a score – compliance teams and reviewers required understandable reasons for escalating decisions which have consistent documentation attached to them (Oztas et al., 2024; Adadi and Berrada, 2018; Arner, Barberis and Buckley, 2017).

2. Research Question

To what extent can cost sensitive, explainable machine learning can improve AML transaction monitoring in terms of improved detection of laundered activity and fewer false positives given a fixed alert capacity when compared with a typology-based rules baseline on the IBM Transactions for Anti-Money Laundering HI-Small data set?

Additionally, is it possible to convert SHAP explanations into regulator friendly “reason codes” that supports an investigation or audit trail but do not detrimentally affect model performance?

3. Justification

Feasible: Method will be created in a form of research prototype with use of public IBM AML (HI-Small) dataset and standard python tools (pandas, scikit learn, XGBoost, SHAP) (Lundberg and Lee, 2017). The pipeline will be designed to run on a typical resource of laptop with features such as memory-aware processing like chunking and saving the intermediate feature tables.

Measurable: Measures of performance based on metrics suitable for severe class imbalance which will be used. Although this will include Precision-Recall AUC and operational measures that capture real capacity of investigation such as recall in the top 0.5%, 1% and 5% highest risk alerts and false positive per true positive (Oztas et al., 2024). At conclusion, results will be compared to both a typology-based rules engine and a logistic regression base-line.

Significant: AML monitoring is one of the fundamental RegTech capabilities (Arner, Barberis and Buckley, 2017). Any reduction in the number of unnecessary alerts can have the direct impact on reducing manual review workload and helping the teams consider cases which are higher risks that will help to improve both efficiency and effectiveness.

Ethical: The dataset for this research is synthetic and publicly available which is away from the privacy concerns of actual customers record. The focus of the study will be transparency through codes of reason and will cover practical trade-offs between being able to catch more suspicious activity and not causing excessively escalation of legitimate transactions (Adadi and Berrada, 2018).

4. Specific Items to be Addressed

Dataset and problem set up: Firstly, this thesis will use the IBM AML HI-Small dataset and “Is_Laundering” as the label target as it is provided within the dataset. Secondly, will set train/validation/test set using Timestamp in order to mitigate look ahead of bias and make sure features behaviour are computed using past only rolling windows in order to avoid the leakage.

Dataset Link: <https://www.kaggle.com/datasets/ealtman2019/ibm-transactions-for-anti-money-laundering-aml>

Feature engineering: Creates behavioral features which is unique in AML patterns such as velocity number/sums within time windows and structuring proxies for example; repeated close to threshold amount which all only can be calculated from information up to time t.

Baselines: Thoroughly deploy a rules engine built on a typology scheme for instance; velocity, structuring, fan-in, fan-out heuristics and calibrate the system against matched alert volumes which is top-K alerts for fair comparisons and will train Logistic Regression as a simple ML benchmark (Oztas et al., 2024).

Cost-sensitive modelling: Along with Logistic Regression model implementation for better ML training will be XG-Boost which will be a cost sensitive method that will train model using class weighting (e.g., scaleposeweight) which to be compared with standard training and resampling with Smote to test if the modelling method is robust to imbalance.

Explainability and Reason codes: Applying SHAP method to deliver local explanations regarding escalated alerts then after that create a level of translation mapping important SHAP drivers into short and consistent reason codes (e.g., High transaction frequency within 1 hour) (Lundberg and Lee, 2017).

Evaluation: PR-AUC report and prioritizing functional operational success at fixed alert capacities including false positives per true positive (Oztas et al., 2024). Additionally adding the slice/error analysis on typology proxies (i.e., high velocity segment vs near-threshold segment) to know where the system works well or fails.

Artefact: Document Jupyter Notebook pipeline that ingests raw transactions to produces a ranked list of alerts with model and its scores also produces a simple STR decision support excel (csv) with selected decision threshold and top reason codes for each escalated alert.

5. Bibliography

Oztas, B., Cetinkaya, D., Adedoyin, F., Budka, M., Aksu, G. and Dogan, H. (2024) 'Transaction monitoring in anti-money laundering: A qualitative analysis and points of view from industry', *Future Generation Computer Systems*, 159, pp. 161–171. <https://doi.org/10.1016/j.future.2024.05.027>

Adadi, A. and Berrada, M. (2018) 'Peeking inside the black-box: a survey on explainable artificial intelligence (XAI)', *IEEE Access*, 6, pp. 52138–52160.

Arner, D.W., Barberis, J. and Buckley, R.P. (2017) 'FinTech, RegTech, and the reconceptualization of financial regulation', *Northwestern Journal of International Law & Business*, 37(3), pp. 371–413.

Lundberg, S.M. and Lee, S.-I. (2017) 'A unified approach to interpreting model predictions', *Proceedings of the 31st International Conference on Neural Information Processing Systems (NeurIPS 2017)*, pp. 4765–4774.